

Feature-Reusing Local Keypoint Refinement for Monocular Pallet Pose Estimation: Frozen Estimator Applications

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Abstract—Accurate image landmarks are important to monocular pallet pose measurement, but improving localization need not require changing an established detector. We study a fixed, synthetic-supervised local keypoint refiner that reuses two feature scales of a frozen RGB estimator. In the completed YOLO application, the 18,962-parameter module predicts a distribution over local corner displacements while preserving bounding boxes, confidence scores, instance selection, and the center landmark. For this application, on 319 reused development images from 13 sessions, the mean over three independently trained seeds of pooled landmark median error changes from 6.616 to 5.905 pixels. The paired-session interval for the reduction is [0.411,1.174] pixels. Complete-denominator keypoint accuracy, error tails, geometry-reference pose errors, and desktop latency are reported separately. A matched-evidence direct regression control and a source-derived PoseFix pallet adaptation test distinct refinement alternatives. The P-prior median difference is 0.336 pixels (paired-session interval [0.063,0.585]). The interval favors the prior, not P. This is a fixed-budget development comparison. These measurements establish the scope of the development evidence, not an unseen-session guarantee. Independent real-session confirmation is not yet available. The extension applies the same correction principle to a separately frozen DOPE estimator with new source-trained heads and explicit missing-point accounting. For frozen DOPE on reused DEV, the conditional keypoint median is 12.585 pixels before refinement and 7.461 pixels for the mean of three P seed statistics; full-GT PCK10 is 19.872 and 36.775 percent, respectively. For the third, dimension-conditioned ResNet18 estimator, the primary P0 refiner changes reused-DEV conditional median error from 8.001 to 7.143 pixels and full-denominator PCK10 from 54.68% to 58.28%; reconstructed-reference translation and canonical camera-facing-frame C2 rotation medians change from 11.007 cm/3.252 deg to 10.001 cm/3.018 deg. Neither a universal model-agnostic result, an original expectation operator, nor operational handling reliability is claimed.

Index Terms—Monocular vision, pallet pose, keypoint refinement, synthetic supervision, feature reuse, measurement evaluation.

I. Introduction

Pallet localization provides geometric input to inspection, tracking, and material-handling systems. A monocular RGB camera offers a compact sensing interface, but image-level detection is not the same quantity as accurate geometric measurement. The same correctly detected instance can have displaced corners; a downstream perspective-n-point (PnP) solver can amplify some image

errors depending on geometry and viewing configuration. It is therefore useful to distinguish detection, conditional landmark precision, and pose measurement when evaluating a refinement method.

Synthetic training can supply image supervision without manually annotating every pallet training image. Synthetic RGB pallet pose estimation is already established, for example through the Euro-pallet study of Knitt et al. [1]. We do not claim its invention, a measured reduction in annotation labor, or a universal synthetic-to-real solution. Instead, we ask a narrower question: can a small local module improve the residual geometry of a fixed estimator while leaving its detection outputs unchanged?

We investigate the already-trained point refiner P. It reads existing detector features and scores candidate displacements around each predicted corner. The detector is frozen, but this is not a guarantee that corrected geometry is harmless: a wrong displacement can still degrade a keypoint or a pose. Our evaluation therefore retains tails, individual sessions, and complete ground-truth denominators alongside the central error statistic. The original application depends on the YOLO feature interface. A second, separately specified DOPE application tests whether the same local readout principle remains useful with different frozen features and error patterns. It requires a new adapter and new heads, not zero-shot transfer of the YOLO weights. A finite set of fixed applications cannot establish universal model-agnostic behavior.

The third application is a completed full-image, nine-landmark Simple-Baselines-derived ResNet18. Its frozen FULL estimator receives RGB and registered physical dimensions, and its primary transfer test adds the same image-feature local refiner without a second dimension input (P0). D0 is the matched direct-residual control. P5 and P5_CONSTANT form a separate dimension-input extension. All real-image outcomes are reused-development evidence.

This revision adds a completed ResNet18 estimator that receives RGB and registered pallet dimensions during training and inference. The historical YOLO and DOPE estimators and their primary P heads remain image-feature based; their physical dimensions still enter only downstream PnP. Within ResNet18, the primary third-backbone transfer arm P0 reads frozen image features and predicted geometry without a second dimension input. A separate P5-P5_CONSTANT experiment tests whether

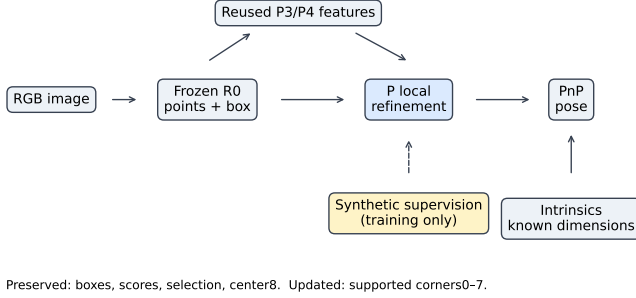


Fig. 1. Information paths of the YOLO and DOPE applications. Their P heads use image features and predicted points/box, while dimensions enter downstream PnP. The ResNet18 FULL estimator separately injects five normalized dimension values into its decoder; primary P0 then refines frozen image features without an additional dimension input. Boxes, scores, selected instance, and center remain fixed within each refiner application.

adding the same five-value dimension context to the correction head contributes beyond the already dimension-conditioned FULL estimator.

The contributions are an explicitly specified feature-reusing refinement interface, within-estimator comparisons of the local distribution readout with a direct residual control, and an auditable protocol tying original-pixel outputs to failure-aware geometry and measured cost. Historical YOLO line-structured and prior-derived comparisons remain visible. Separate DOPE and ResNet18 applications test the same correction principle through their own adapters and newly trained heads; the ResNet18 dimension experiments remain a distinct extension. These are system and empirical contributions to be assessed against the evidence. We do not introduce a new expectation operator, a low-rank adaptation algorithm, or a new safety controller. Figure 1 separates the historical learned correction path from downstream geometry inputs.

II. Related Work and Comparison Scope

A. Pallet perception and synthetic supervision

DOPE[2] estimates semantic cuboid landmarks from RGB. Knitt et al. [1] trained a DOPE-based pallet estimator using synthetic imagery and evaluated RGB Euro-pallet pose. More recently, Miura et al. [3] studied monocular pallet pose using metric-depth predictions and ground-plane information. These works motivate the application and establish that RGB pallet estimation and synthetic data are not new in themselves. Their datasets and auxiliary information differ from ours; published numbers are not subtracted from the present development results.

B. Pose and coordinate refinement

PoseFix[4] refines an input pose using the image and a separate backbone. Its human-pose training procedure synthesizes pose errors from human-keypoint statistics.

Our purpose adaptation replaces those human assumptions with frozen source-training predictions from the pallet detector. It is named PoseFix-derived pallet9, matched-exposure adaptation, not a reproduction of the original human benchmark. Integral regression[5] establishes differentiable coordinate expectation as prior art. P applies an expectation to candidate displacements; the expectation itself is not a claimed invention.

CRT-6D[6] uses object-surface keypoint features in cascaded pose refinement. It is relevant to sparse feature-based correction, but has a different object-surface and pose contract. The inspected official repository at the recorded commit contains no executable network; no substitute transformer is reported as CRT-6D. Our named prior experiment is limited to the declared PoseFix-derived family rather than expanding a benchmark after seeing results.

C. Feature reuse and small side networks

Side-tuning[7] relates to adapting a fixed representation through a separate learned path. P shares the general motivation of retaining a backbone and training a smaller addition. Nevertheless, P does not implement LoRA, and no new parameter-efficient fine-tuning benchmark is established. Both P and the direct control D have small learned heads. Their difference therefore cannot be attributed simply to parameter efficiency. Freezing the backbone also does not make inference overhead zero.

III. Fixed Refinement Method

A. Sensor, detector, and coordinates

The historical baseline R0 is a YOLO26n pose estimator producing a box, confidence, and nine points per candidate instance. The selected instance is the highest-confidence prediction under the existing pipeline. Corner indices 0–7 retain the registered cuboid convention, and index 8 retains the established center definition. The wrapper does not use a ground-truth box or pose to select the input instance. Ground truth is used downstream for training supervision and evaluation matching only.

R0 was initialized from a pretrained YOLO26n pose checkpoint, and its stored provenance identifies COCO-pose training in that checkpoint. The pallet run also records pretrained initialization. Thus, “synthetic-supervised refinement” refers to the training of the additional module, not a claim that no real annotations ever contributed to the full estimator. Real labels are also used in development and evaluation. R0 is not retrained in the present comparison.

The historical image pipeline is preserved: rectangular detector preprocessing at the registered input size, stride-aware padding, and original-image restoration. Prepared synthetic images already include their recorded reflection padding and are not padded a second time. Let A be the diagonal image-to-network scaling matrix and $\mathbf{o}$ its offset. Original points $\mathbf{p}$ and network-input points $\tilde{\mathbf{p}}$ satisfy $\tilde{\mathbf{p}} = A\mathbf{p} + \mathbf{o}$. YOLO letterboxing gives $A = gI$; the DOPE raster

can have unequal actual horizontal and vertical scales. Corrections use A^{-1} on displacements and add them to the retained original points. A zero correction therefore preserves the exact baseline coordinates.

B. Local candidate evidence

P adapts P3/P4 features with small learned channel projections. Around each valid corner, 13 directions and 17 radii define 221 displacement candidates within 0.08 predicted-box diagonal. A zero-displacement null candidate completes the 222-way distribution. Each non-null candidate reads a 4×8 stencil from both feature scales using the registered bilinear sampling geometry. A shared patch encoder and corner-role/context features feed candidate and null scorers. No explicit edge or Hough representation is required by P.

Let d_b denote the predicted box diagonal in network-input pixels, $\mathbf{v}_j$ the normalized candidate offsets, and z_{kj} the score for corner k . For the frozen temperature T and scale λ , the proposed displacement is

$$\delta_k = \lambda d_b \sum_j \frac{\exp(z_{kj}/T)}{\sum_l \exp(z_{kl}/T)} \mathbf{v}_j. \quad (1)$$

The null candidate has $\mathbf{v}_{\text{null}} = \mathbf{0}$. Given an original-image diagonal d_I and frozen cap fraction α , movement and restoration are

$$\Delta_k = A^{-1} \delta_k, \quad (2)$$

$$\bar{\Delta}_k = \Delta_k \min \left(1, \frac{\alpha d_I}{\|\Delta_k\|_2} \right), \quad (3)$$

$$\mathbf{p}'_k = \mathbf{p}_k + \bar{\Delta}_k, \quad (4)$$

with unit scaling for a zero movement and no clipping when the selected cap is absent. Historical YOLO P retains its selected $\lambda = 1$, $\alpha = 0.01$, and per-seed temperatures. DOPE uses the same source-only selection grid; its selected values are supplied with its own result records. Unsupported points are unchanged; point 8 is copied exactly. Detection boxes, confidences, candidate order, and nonselected points are preserved. These contracts do not imply unchanged or always improved 6D geometry.

C. Training and internal controls

P learns a soft distribution over candidate displacements from synthetic corner residuals. The finite target support and per-frame reduction are retained from the fixed implementation. The historical YOLO P, D, and L fits each use their recorded three seeds and fixed exposure budget; those models are not retrained. YOLO P has 18,962 learned parameters. D reuses the same sampling and shared feature descriptors, but regresses normalized corner residuals with an L1 objective. The YOLO D instance has 19,450 parameters and an unbounded raw output, in contrast to P's candidate convex hull. This is a comparison of readout/supervision packages, not a causal ablation of softmax alone.

L is the historical line-structured internal control. Its results are retained to show that explicit line structure has

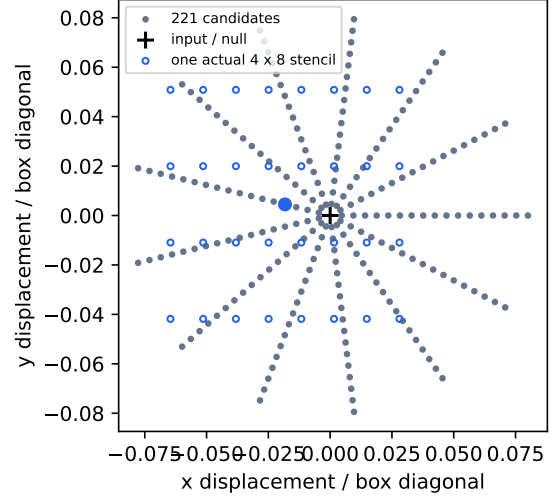


Fig. 2. The implemented candidate lattice and one actual sampling stencil in box-diagonal units. The blue stencil is shown at a fixed candidate index, not selected for performance. The probability expectation in (1) combines candidate displacements; no illustrative weights are presented as measured outputs. Geometry and source hashes are recorded in `FIGURE_MANIFEST.json`.

not established an all-metric benefit in this configuration. Earlier Hough, active-adaptation, and replay investigations are part of the research history, not new primary comparisons in this manuscript. No unfavorable historical outcome was deleted.

For a supported corner with target residual $\mathbf{r}_k$, P's target mass is

$$q_{kj} = \frac{\exp[-\|d_b \mathbf{v}_j - \mathbf{r}_k\|^2 / (2\sigma^2)]}{\sum_l \exp[-\|d_b \mathbf{v}_l - \mathbf{r}_k\|^2 / (2\sigma^2)]}, \quad \sigma = 0.08 d_b / 17. \quad (5)$$

Cross-entropy is averaged over supported corners within each frame and then over frames with support. This smooth finite-candidate target differs from D's direct residual L1 supervision. Its existing settings are retained, not newly optimized here.

D. Second fixed estimator: DOPE adapter

The second application freezes an existing six-stage pallet DOPE checkpoint and changes the feature adapter rather than the correction principle. The checkpoint is identified by SHA-256 in the protocol; no DOPE weight is updated by the refiner fits. We use the existing semantic-channel local-peak decoder at threshold 0.3. This implementation does not run affinity-based multi-object grouping. Its fixed box is the hull of at least three valid predicted corners, and its score is the maximum corner belief. Thus, this is an application to the declared single-pallet DOPE pipeline, not a reproduction of every original DOPE inference option or a detector-ranking experiment.

The two frozen VGG taps are post-ReLU layers 17 and 26, with 256 channels at stride 4 and 128 at stride 8. Learned projections map both to the same 16-channel evidence interface used by P and D; the shared encoded width

is 24. The candidate lattice, stencil geometry, corner ordering, null candidate, losses, and displacement restoration remain as specified above. Input-channel counts change the number of learned parameters; the YOLO count is not assigned to the DOPE heads. A matched-evidence D control is trained for each DOPE seed. L and the separate PoseFix-derived comparator are historical YOLO controls and are not silently presented as new DOPE experiments.

The fixed adapter applies reflection padding of 100 pixels to native real RGB, resizes the padded canvas back to the native size, then uses the recorded shortest-side-400, multiple-of-eight raster. Synthetic source RGB already contains this reflection preparation and receives no second padding. The exact composed affine, including actual raster widths and heights, maps predictions back to the caller’s coordinates. Training retains the source preparation; DEV restores unpadded original pixels before measurement. The inference preprocessing and normalization are checked against the selected existing inference path, not claimed to be byte-identical to the original augmented DOPE training pipeline.

Predicted missing points stay missing. P and D cannot create a previously absent corner, change the fixed hull or score, or replace the center landmark. A point without the required feature support is copied rather than supplied with a target-derived value. These invariants isolate local refinement but also bound what it can repair: low coverage cannot be hidden by reporting only an accurate surviving subset. Camera intrinsics and physical dimensions remain exclusive to downstream PnP.

E. Third application: full-image ResNet18

The third estimator is a full-image pallet adaptation of Simple Baselines[8], using an ImageNet-initialized ResNet18 body, three 256-channel stride-two deconvolutions, and nine stride-four spatial distributions. It uses no YOLO detection or ground-truth crop. Prepared RGB is aspect-fitted to 384×512 with centered letterboxing and ImageNet FP32 normalization. A spatial softmax expectation decodes all nine landmarks. The objective is normalized Gaussian spatial cross-entropy weighted by 0.1 plus smooth-L1 coordinate loss; this replaces the earlier unusable masked-MSE pilot and is the only ResNet baseline used in the reported results.

Canonical (W, D, H) is converted to five TRAIN-normalized log values: $\log W$, $\log D$, $\log H$, $\log(W/D)$ and $\log(H/\sqrt{WD})$. FULL applies all five through a decoder FiLM block, SHAPE keeps the two ratios, and CONSTANT supplies zeros through the same architecture. The completed direct-estimator study trains CONSTANT, SHAPE and FULL arms for a fixed 10 epochs on 55,980 synthetic TRAIN images (34,990 updates and 559,800 exposures per arm). All share seed 42, ImageNet ResNet18 initialization, identical source order and photometric augmentation. FULL supplies normalized $\log W$, $\log D$, $\log H$, $\log(W/D)$ and $\log(H/\sqrt{WD})$ to a decoder FiLM block; SHAPE keeps only the two ratios and

CONSTANT supplies zeros. Epoch 10 is fixed using no real-image selection. After FULL is frozen, D0 and P0 read layer2/layer3 features (128/stride8 and 256/stride16). P0 is the primary third-backbone application of the same 222-candidate local distribution readout; D0 is its direct-residual control. P5 adds a zero-initialized dimension-conditioned logit residual, while P5_CONSTANT keeps the identical structure but zeros that context. Every head is independently initialized; no YOLO or DOPE head is a warm start.

F. Purpose-adapted prior comparator

The historical YOLO prior uses the official PoseFix ResNet152 arrangement, an aspect-corrected and 1.25-expanded predicted-box RGB crop of 384×288 , and nine Gaussian input pose maps. It adds ImageNet TF-Slim initialization, masks center loss and restores the original center. Human swap/inversion noise is replaced by actual frozen source-detector errors. Operator and training parity checks for the source-based PyTorch port, including the effective fused-BN epsilon and target-boundary handling, are detailed in the supplement.

Its declared budget is three seeds of 6,000 Adam updates at batch 16 in the exact historical P sample order, using the final checkpoint. This is a matched-exposure adaptation rather than the original 140-epoch human-pose protocol or a claim of convergence. All three matched-budget fits were completed; the learning curves are diagnostic, not a best-checkpoint selector.

G. Downstream geometric measurement

The unchanged prediction-only selector uses the registered camera intrinsics and physical pallet dimensions to select its dimension hypothesis. The selected cuboid is solved using the canonical SQnP/RefineLM path. Intrinsics are never neural inputs. Dimensions are not neural inputs to the historical YOLO/DOPE P/D heads or ResNet18 P0/D0, but they are inputs to the ResNet18 FULL estimator and its separate P5 head extension. Ground truth does not choose the prediction’s most favorable pose hypothesis. Current reference poses are reconstructed from manual image points and known geometry, rather than obtained from an independent physical 6D instrument. Near-square axis confusion is not newly excused by adding a symmetry minimization.

IV. Datasets and Measurement Protocol

A. Source exposure and rule selection

For historical YOLO, the source manifest contains 55,980 training records, with 55,915 usable matched training rows, and calibration/selection/heldout partitions of 1,004/1,031/1,985 records. Source images, labels, row order, and completed model choices are hash-bound. Actual RGB crops, rather than guessed images reconstructed from cached features, are used for the prior backbone. Synthetic source subsets are historically reused and are not described as universally untouched data.

TABLE I

Added inference inputs. All added heads use zero real-pallet training examples. Dimensions are a neural input only for ResNet18 FULL and the separate P5 extension.

Method	Added inference input	Added pretrain-ing
R0	none	none
P/D/L	R0 P3/P4, pose/box	none
Prior	RGB crop, pose/box	ImageNet classes
DOPE P/D	VGG taps, pose/box	none
ResNet FULL	RGB, five dimensions	ImageNet classes
ResNet P0/D0	FULL layer2/3, pose/box	none
ResNet P5	P0 evidence, five dimensions	none

The prior shares the existing synthetic-only lambda/cap grid, objective, and tie rule. Coordinate outputs do not receive a fabricated temperature sweep. Raw prior and capped-wrapper results are reported separately. Heldout results are computed after selection; DEV does not choose the checkpoint, learning rate, noise model, or wrapper. The historical model panel is frozen before any independent confirmation annotations may be opened; the DOPE extension also freezes source-selected rules before its DEV replay.

B. Fixed DOPE source budget and selection

DOPE uses the same 60,000-record synthetic RGB source manifest: 55,980 training, 1,004 calibration, 1,031 selection, and 1,985 heldout records. A usable training row needs the frozen predicted hull to meet IoU 0.5 with the source object and at least one jointly valid corner. Its usable count is estimator-specific and must not inherit the historical YOLO count. Each of three seeds trains a fresh P and a fresh D for exactly 6,000 AdamW updates of nominal batch 16. Within a seed they share the exact row sequence and frozen feature evidence, giving 96,000 nominal source exposures per head. Cross-estimator usable rows need not coincide. Frozen VGG prefixes run online; the DOPE stage heads are not rerun inside these training steps. Feature maps use the original FP16 roundtrip, with FP32 heads and no mixed-precision head training.

The learning-rate, warmup, decay, gradient clipping, and final-checkpoint rule are fixed in advance; no DEV-based early stopping or architectural sweep is allowed. A discarded two-update smoke check is a separate implementation check, not a seventh reported fit or a warm start. P temperature is selected on the calibration partition from $\{0.5, 1, 2, 4\}$ using supported-frame target cross-entropy. D has no temperature selection. Each family selects one three-seed mean rule on the selection partition from $\lambda \in \{0, 0.0625, 0.25, 1, 4\}$ and cap fractions $\{\text{none}, 0.01\}$. The fixed tie rule favors smaller λ and then the stricter cap. Missing source supervision contributes the declared failure penalty; no real image chooses a wrapper. Heldout measurements follow rule freezing and remain historically reused source data.

The baseline-training budgets are not matched: the DOPE checkpoint ending in epoch 0060 was produced by

the inclusive 0–60 loop, whereas YOLO uses its selected best.pt. Shared source imagery and matched P/D head exposure do not establish equal baseline optimization, augmentation, pretraining, or compute. The supplement records this distinction alongside the source and checkpoint hashes.

C. Fixed ResNet18 source budget and selection

CONSTANT, SHAPE and FULL each train for the fixed 10 epochs on all 55,980 synthetic TRAIN images, batch 16, for 34,990 updates and 559,800 image exposures. The three arms share initialization, source order and augmentation; only the dimension mask changes. Adam starts at 10^{-3} with zero weight decay and factor-ten drops at epochs 7 and 9. Synthetic calibration is monitored after each epoch, but epoch 10 is fixed and no real result selects the checkpoint.

FULL is then frozen. Each D0/P0/P5/P5_CONSTANT head trains for 6,000 AdamW updates at batch 16 in each of three seeds, using 55,804 usable TRAIN rows and identical within-seed order. Calibration temperature and movement rules are fixed on synthetic calibration/selection, and source heldout is opened afterward. P0 is the transfer experiment used for the third-backbone claim. P5 is a separate input extension; comparing P5 with P5_CONSTANT isolates the incremental five-value input to that head, while P5–P0 also changes path and capacity.

D. Development population and supervision

The historical YOLO positive development panel has 319 images in 13 recorded sessions; 2689 negative images accompany the detection-preservation audit. In that historical panel there are 311 matched positive frames and 2,756 supervised points for conditional precision, versus 2,818 supervised points in the complete ground-truth denominator. Point supervision is the original label contract and is not renamed as measured physical visibility. Visible boundaries, inferred amodal corners, and unavailable landmarks require distinct interpretation in new annotation QA.

The DOPE extension reuses the identical 319 positive images and 2,818 supervised points, but reports its own matched and missing counts. Canonical conditional precision requires the fixed box match and all nine predicted points finite; a separate partial-point diagnostic cannot replace it. P and D preserve this baseline support within DOPE, without requiring equal support across estimators. There is no new DOPE negative-image/AP experiment.

ResNet18 also reuses these 319 images and 2,818 landmarks. Its full-image direct estimator always emits one nine-point pallet hypothesis, so predicted-hull matching is not detector recall. The primary refiner evaluation fixes FULL support at 299 matched frames and retains all 319 rows for PCK and pose coverage. Plastic and wood are reported separately. Wood $0.80 \times 0.59 \times 0.14$ m has depth

below the synthetic TRAIN minimum and is dimension-support extrapolation. GREEN150 and the 0918 set of 119 square pallets are 2D-only development panels whose registered dimensions are fixed at $1.10 \times 1.10 \times 0.15$ m within each set.

The primary statistic pools supervised original-pixel Euclidean errors within each model seed, computes its median, and then averages the three seed statistics. It is neither an ensemble prediction nor the average of frame medians. Its conditional precision denominator excludes unmatched frames, which is why complete-denominator PCK is also required. For threshold t ,

$$\text{PCK}_t = \frac{\sum_{i,k} h_{ik}(t)}{N_{\text{GT}}}, \quad (6)$$

where missing predictions, unmatched top instances, and nonfinite points contribute zero hits, not zero error. Matching remains at the registered box IoU threshold. If any support differs, an intersection-only median cannot replace this failure-aware report. In the DOPE extension a conditional paired interval is not computed when its per-frame support differs; the mismatch is reported alongside full-denominator PCK and pose coverage.

Registered object sizes and the per-annotation camera intrinsics are tabulated in the supplement. They are inputs to this geometry-reference evaluation, not newly certified metrology. The canonical solver uses no distortion coefficients; independent camera calibration uncertainty is not established by these annotation records. The declared long-axis benchmark admits a half turn but not a quarter turn. In particular, the wood object’s visual symmetry under a half turn is unverified. Such conventions do not establish physical fork-entry interchangeability.

E. Uncertainty and pose metrics

We use 10,000 paired session bootstrap draws with seed 20260914, applying each draw’s frame/session multiplicities to every model seed. The P–R0 statistic compares against one baseline, not three copies treated as independent models. P–D and P–prior are secondary development contrasts. Secondary intervals are not multiplicity-adjusted and are not confirmatory superiority, equivalence, or noninferiority tests. The reused development data cannot be made independent by specifying this protocol after previous analyses.

Rotation, yaw, translation, 3D IoU, and proper-group C2-aware corresponding-point ADD AUC are evaluated under the existing geometry-reference convention. The ADD metric is not unrestricted nearest-neighbor ADD-S. Pose coverage and 2D matching are different denominators: a solver can return a pose from an incorrectly matched top detection. Translation-error changes in centimeters can be expressed in millimeters, but a pixel reduction cannot be converted to one universal millimeter gain. No insertion trials or independently instrumented physical pose measurements are included.

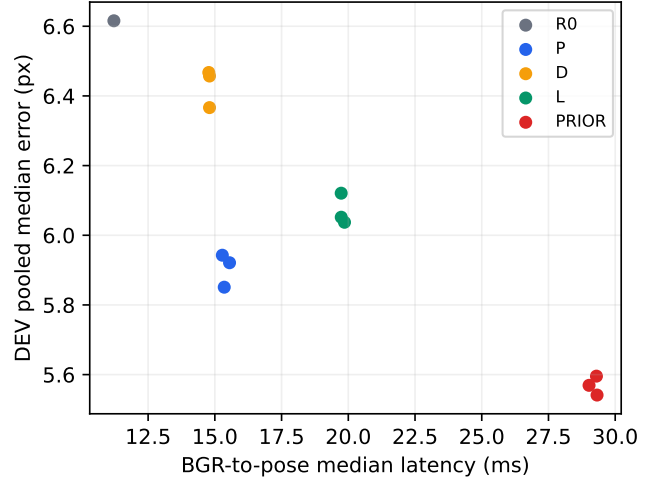


Fig. 3. Historical YOLO per-seed DEV median against same-session median full-path latency. Lower is preferable on both axes. Each seed is a point, not a timing-selected winner or ensemble; separate-backbone prior cost is included. Source result/runtime hashes are recorded in FIGURE_MANIFEST.json.

F. Cost and independent confirmation

The expanded panel was measured in the same session for all comparators.

The integrated desktop timing starts from the same 26 native decoded BGR image bytes for every estimator. It includes each model’s preprocessing, 2D inference/refinement and restoration; canonical prediction-only PnP is measured separately and also included in the full path. Loading and file decoding are excluded. Every arm uses batch one, 20 warmups and five balanced cyclic/reversed repeats with Torch intra/inter-op and OpenCV threads fixed to one. Seed 1 is the prespecified learned-head representative. All 1,170 measured rows are retained, including DOPE pose failures; no fastest repeat is selected. Saved coordinate and pose replays bound maximum differences to 2.65×10^{-4} pixels and 2×10^{-4} in recorded pose metrics. This desktop RTX 3080 panel does not estimate Jetson latency, energy, concurrent throughput or training cost.

Independent confirmation is awaiting new data. The collection plan starts with 12 separate sessions and 15 prespecified frames per session, with at least 30 frames independently double annotated before blind adjudication. This is not a power guarantee. Predictions are hidden during annotation and the full panel must be frozen before evaluation. Existing FINAL-named manifests have unavailable membership and are not treated as independent observations.

V. Results

A. Completed historical YOLO comparisons

Table II summarizes the original-pixel and geometry-reference measurements. Complete per-seed tables, extra PCK thresholds, coverage, and uncertainty are supplied

TABLE II

Reused DEV: seed-mean statistics, not an ensemble. Translation is against the geometry reference. NM means not measured in this build.

Model	Median px	P90 px	PCK10 %	Trans. cm
R0	6.616	38.670	63.73	7.897
P mean	5.905	37.766	67.34	7.153
D mean	6.430	37.566	64.57	7.597
L mean	6.070	37.320	66.78	7.548
PRIOR mean	5.569	37.227	68.41	6.895

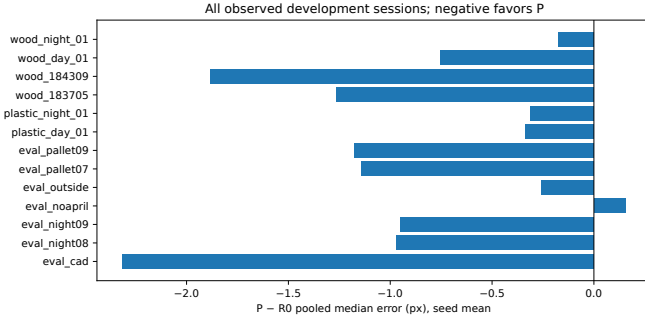


Fig. 4. P-R0 development median effects by recorded session. Negative values favor P. These are descriptive effects from the reused panel, not independent confirmation intervals. All sessions are retained.

in the supplement. P changes the baseline pooled median from 6.616 to 5.905 pixels, a reduction of 0.711 pixels. Its paired-session reduction interval is [0.411, 1.174] pixels. PCK10 changes from 63.733% to 67.341%. These figures describe the present development panel only.

The P-D median difference is -0.525 pixels, with paired-session interval [-0.980, -0.210]. This supports a lower central development residual for P than this fixed direct control. It does not prove that direct regression as a method class is inferior. D has a slightly lower observed P90, and the P-D tail interval includes zero. D also has lower measured desktop latency. There is therefore no all-metric dominance claim. The internal line control also has a lower observed P90 than P.

Figure 4 retains every recorded session. The eval_noapril session has an adverse P-R0 central effect; the other twelve observed sessions improve. Leave-one-session-out estimates remain favorable in this panel, but such deletion diagnostics are not new independent datasets. No unfavorable session is removed from the aggregate.

The seed-mean reduction in geometry-reference translation median is 7.439 mm. This is a change in error statistics against a reconstructed reference, not measured insertion improvement. It cannot establish physical metrology accuracy independently of the image-label and reference-construction uncertainty.

B. Prior-derived comparison

The prior checkpoints were fixed at the final training update; only the wrapper rule was selected on synthetic data. The selected wrapper has lambda 1.000 and cap fraction 0.01. Raw and wrapped outputs are separate supplementary rows. All wrapped prior seeds retain the baseline matched supervision support, with no nonfinite supervised coordinates. The P-prior paired median difference is 0.336 px, with session interval [0.063, 0.585] px. This interval favors the prior, so P is not the most accurate comparator under this central-error criterion. A negative contrast favors P; a positive contrast favors the prior. The complete-denominator PCK, tail and pose rows must be considered alongside the median. Six thousand updates per seed do not certify saturation or reproduce the original 140-epoch protocol.

C. Second-estimator development evidence

For frozen DOPE on reused DEV, the conditional keypoint median is 12.585 pixels before refinement and 7.461 pixels for the mean of three P seed statistics; full-GT PCK10 is 19.872 and 36.775 percent, respectively. The tables retain missing predictions and unfavorable outcomes.

The supplement reports the full per-seed table, paired intervals, source-heldout statistics and selected wrappers. Within DOPE, central residuals, complete-denominator accuracy, tails, and pose coverage are separate outcomes. A lower conditional median alone cannot establish a net sensing improvement when missing predictions remain. Across estimators, unequal baseline errors, boxes, features, and upstream training prevent a causal attribution to backbone identity. Any favorable DOPE effect supports this fixed application only; a null or adverse result remains part of the same experiment.

D. Third-estimator development evidence

The ResNet18 FULL estimator and all twelve D0/P0/P5/P5_CONSTANT head fits completed. P0 is the primary third-backbone transfer arm; it improves central 2D development error and PCK10, while its translation interval includes zero. P5 is reported only as a separate dimension-conditioned-head extension because P5-P5_CONSTANT does not establish stable joint 2D/T/R improvement. For primary P0-FULL, the paired session-bootstrap change in conditional median is -0.859 pixels [95% CI -1.318, -0.621] and PCK10 changes by +3.596 percentage points [2.515, 4.986]. Canonical MAIN camera-facing-frame C2 rotation changes by -0.234° [-0.502, -0.022], while translation changes by -1.007 cm [-1.668, 0.187] and therefore retains zero in its interval. Figure 5 shows all three primary and dimension-extension point estimates; individual seeds and intervals are in the supplement.

The separate direct-input ablation, scored in its physical-frame proper-group convention, does not show stable dimension benefit: FULL versus CONSTANT

TABLE III

New fixed DOPE application on reused DEV. D/P rows average three per-seed statistics, not predictions. Keypoint medians are conditional; PCK retains 2818 GT points. Pose medians are geometry-reference errors on available fits; coverage uses 319 frames. No cross-estimator equality of matched support is assumed.

Arm	Med. px	P90 px	PCK10 %	Match/319	T cm	R deg	Pose %
DOPE	12.585	50.609	19.872	190.0	10.046	3.441	65.831
DOPE+D mean	7.867	48.487	36.054	190.0	8.323	3.018	65.831
DOPE+P mean	7.461	50.713	36.775	190.0	8.529	2.884	65.831

TABLE IV

Primary ResNet18 frozen-estimator transfer on reused DEV319. FULL is the dimension-conditioned direct estimator; D0 and P0 are means of three per-seed statistics, not prediction ensembles. P0 is the primary local-refinement arm and has no additional dimension input. Conditional 2D errors require the fixed baseline hull match; PCK10 retains all 2,818 supervised landmarks. Pose errors use the reconstructed geometry reference, R is canonical MAIN camera-facing-frame C2 rotation, and all arms have 319/319 solver coverage.

Arm	Med. px	P90 px	PCK10 %	Match/319	T cm	R-C2 deg	C2-aware ADD AUC
FULL baseline	8.001	52.054	54.68	299	11.007	3.252	0.31296
D0 mean	7.852	51.928	55.26	299	9.837	3.218	0.32604
P0 mean	7.143	52.086	58.28	299	10.001	3.018	0.34219

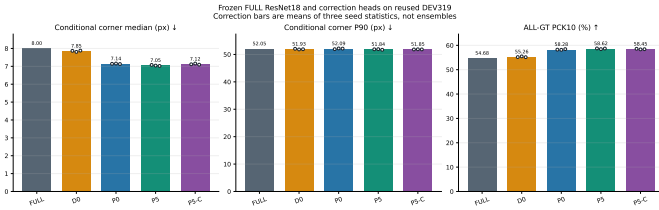


Fig. 5. ResNet18 FULL and correction-head results on reused DEV319. D0/P0/P5/P5-C bars are means of three seed statistics; circles show individual seeds. P0 is the primary third-backbone transfer arm, while P5/P5-C are the separate dimension-input extension.

changes DEV median by -0.210 pixels, reconstructed-reference translation by $+1.269$ cm and rotation by -0.760° . The canonical correction-head comparison P5-P5_CONSTANT is likewise small: -0.074 pixels $[-0.116, 0.019]$, -0.071 cm $[-0.115, 0.071]$ and -0.027° $[-0.067, 0.025]$. Square-set improvements are transfer observations under a constant dimension vector, not evidence that frame-varying dimensions caused the change. Direct physical-frame pose values and canonical refiner pose values are not cross-copied. Cross-estimator comparisons remain descriptive because baseline training, boxes and support differ.

E. Efficiency and accuracy-cost tradeoffs

The expanded panel was measured in the same session for all comparators.

Seed 1 is the prespecified deployment representative and must not be confused with the three-seed accuracy means. In the integrated panel, median full-path time changes from 11.092 to 14.779 ms for YOLO, 64.585 to 67.411 ms for DOPE, and 9.453 to 12.193 ms for ResNet18 FULL+P0. These are measured within-estimator costs. Baseline rankings also reflect different architectures, pre-processing and training; they do not causally rank backbone quality or extrapolate to embedded hardware.

F. Independent confirmation

Not yet measured: no independently reviewed capture/annotation panel is available. The proposed collection count is a plan, not a completed experiment. Development observations are not relabeled as confirmation results.

VI. Limitations and Interpretation

First, the primary development improvement is conditional on matched supervision and on a historically selected model family. Complete-denominator PCK limits one failure mode of optimistic reporting, but does not turn reused DEV into a confirmation set. The session intervals condition on only thirteen observed sessions and three trained seeds; they do not quantify every deployment shift.

Second, the comparison contracts differ in useful ways that must remain visible. P reads internal R0 features; the PoseFix-derived prior computes a separate RGB representation and adds ImageNet pretraining. D has a different output support and loss. Equal source exposure is not equal compute or equal optimization difficulty. None of these comparisons identifies a universal architectural optimum. Bounded prior learning curves diagnose progress but cannot certify full convergence.

Third, current pose errors are measured against a geometry-reconstructed reference. Annotation ambiguity, calibration, object dimensions, and reference construction can affect interpretation. Independently blinded duplicate labels and reviewed capture lineage are required for new confirmation; physical 6D accuracy would require additional metrology. Neither preserved detections nor improved median error guarantees harmless tails, correct pallet axes, or safe handling.

Finally, each application is tied to a declared frozen feature interface and a desktop environment. DOPE and ResNet18 require their own channel/stride adapters, decoders, support rules and new head fits; three completed

TABLE V

Integrated same-image desktop latency (ms), representative seed 1 for learned heads. All methods use the same 26 decoded BGR images, batch one, 20 warmups and five retained repeats. Loading and decoding are excluded; model-specific preprocessing, 2D output and canonical prediction-only PnP are included.

Method	2D med.	2D P90	Full med.	Full P90
YOLO R0	9.554	11.594	11.092	13.184
YOLO+P1	13.174	15.468	14.779	16.959
DOPE	62.894	76.766	64.585	78.870
DOPE+P1	66.291	80.003	67.411	81.361
ResNet18 FULL	7.835	10.432	9.453	12.146
ResNet18+P0-S1	10.520	13.565	12.193	14.982

applications still do not prove arbitrary-backbone transfer. The ResNet dimension results are also bounded: wood is outside source dimension support, both square sets hold dimensions constant, and neither direct FULL-CONSTANT nor P5-P5_CONSTANT establishes stable joint 2D/T/R improvement. Reusing DEV for all three applications does not create independent confirmation. Embedded export parity and Jetson latency are not measured. The work provides an auditable sensing experiment, not a demonstrated control system. The author team must assess whether the remaining independent evidence and scientific contribution are sufficient for submission; compilation and formatting checks are not an acceptance decision.

VII. Conclusion

The completed YOLO application of a fixed feature-reusing point refiner improves the central landmark residual on the reused pallet development panel with a measured desktop cost. The preserved detection interface, failure-aware denominators, internal controls, and source-derived prior comparison make the scope of this claim explicit. The source-derived prior comparison quantifies a fixed-budget tradeoff, not converged-method superiority. Independent real-session confirmation remains required. The DOPE measurements describe one additional fixed application; they do not establish arbitrary-backbone transfer, independent TEST generalization or stable physical T/R improvement. A third frozen feature interface supports the primary P0 refinement result on reused DEV, but the separate P5-P5_CONSTANT comparison does not establish a robust incremental benefit from dimensions or stable physical T/R improvement. Further claims about independent generalization, physical sensing accuracy, or operational success require the corresponding evidence rather than a more favorable presentation of development numbers.

Data, Code, and Author Declarations

Scripts, small numerical artifacts, coordinate checks, source/weight hashes, and reproduction commands are assembled in the companion repository paths identified in the supplement. Raw images, videos, and large external/model weights are not publicly redistributed without permission. The source ImageNet checkpoint retrieval

procedure is documented separately. Author names, affiliations, ORCID identifiers, funding, conflicts, related prior publications, AI-use disclosure, and submission consent remain for the actual author team to confirm. This internal draft has not been submitted.

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